

- Albert Q. Jiang, Alexandre Sablayrolles, Antoine Roux, Arthur Mensch, Blanche Savary, Chris Bamford, Devendra Singh Chaplot, Diego de Las Casas, Emma Bou Hanna, Florian Bressand, Gianna Lengyel, Guillaume Bour, Guillaume Lample, L'elio Renard Lavaud, Lucile Saulnier, Marie-Anne Lachaux, Pierre Stock, Sandeep Subramanian, Sophia Yang, Szymon Antoniak, Teven Le Scao, Théophile Gervet, Thibaut Lavril, Thomas Wang, Timothée Lacroix, and William El Sayed. 2024. [Mixture of experts](#). *ArXiv*, abs/2401.04088.
- Albert Qiaochu Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de Las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, L'elio Renard Lavaud, Marie-Anne Lachaux, Pierre Stock, Teven Le Scao, Thibaut Lavril, Thomas Wang, Timothée Lacroix, and William El Sayed. 2023. [Mistral 7b](#). *ArXiv*, abs/2310.06825.
- Muhammad Khalifa, David Wadden, Emma Strubell, Honglak Lee, Lu Wang, Iz Beltagy, and Hao Peng. 2024. [Source-aware training enables knowledge attribution in language models](#). *ArXiv*, abs/2404.01019.
- Wojciech Kryscinski, Bryan McCann, Caiming Xiong, and Richard Socher. 2019. [Evaluating the factual consistency of abstractive text summarization](#). In *Conference on Empirical Methods in Natural Language Processing*.
- Bar Lanyado. 2023. Can you trust chatgpt's package recommendations? *Vulcan*.
- Jon M Laurent, Joseph D Janizek, Michael Ruzo, Michaela M Hinks, Michael J Hammerling, Siddharth Narayanan, Manvitha Ponnampati, Andrew D White, and Samuel G Rodrigues. 2024. Lab-bench: Measuring capabilities of language models for biology research. *arXiv preprint arXiv:2407.10362*.
- Junyi Li, Xiaoxue Cheng, Xin Zhao, Jian-Yun Nie, and Ji-Rong Wen. 2023. [HaluEval: A large-scale hallucination evaluation benchmark for large language models](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 6449–6464, Singapore. Association for Computational Linguistics.
- Stephanie Lin, Jacob Hilton, and Owain Evans. 2021a. Truthfulqa: Measuring how models mimic human falsehoods. *arXiv preprint arXiv:2109.07958*.
- Stephanie C. Lin, Jacob Hilton, and Owain Evans. 2021b. [Truthfulqa: Measuring how models mimic human falsehoods](#). In *Annual Meeting of the Association for Computational Linguistics*.
- Chris Lu, Cong Lu, Robert Tjarko Lange, Jakob Foerster, Jeff Clune, and David Ha. 2024. The ai scientist: Towards fully automated open-ended scientific discovery. *arXiv preprint arXiv:2408.06292*.
- Alex Troy Mallen, Akari Asai, Victor Zhong, Rajarshi Das, Hannaneh Hajishirzi, and Daniel Khashabi. 2022. [When not to trust language models: Investigating effectiveness of parametric and non-parametric memories](#). In *Annual Meeting of the Association for Computational Linguistics*.
- Potsawee Manakul, Adian Liusie, and Mark John Francis Gales. 2023. [Selfcheckgpt: Zero-resource black-box hallucination detection for generative large language models](#). *ArXiv*, abs/2303.08896.
- Joshua Maynez, Shashi Narayan, Bernd Bohnet, and Ryan McDonald. 2020. [On faithfulness and factuality in abstractive summarization](#). In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 1906–1919, Online. Association for Computational Linguistics.
- Sachin Mehta, Mohammad Hossein, Sekhavat Qingqing, Cao Maxwell, Horton Yanzi, Chenfan Jin, Sun Iman, Mirzadeh Mahyar, Najibi Dmitry, Belenko Peter, Zatloukal Mohammad, and Rastegari Apple. [Openelm: An efficient language model family with open-source training and inference framework](#).
- Meta Llama 3. 2024. Introducing meta llama 3: The most capable openly available llm to date. <https://ai.meta.com/blog/meta-llama-3/>. Accessed: 6/15/2024.
- Sewon Min, Kalpesh Krishna, Xinxin Lyu, Mike Lewis, Wen-tau Yih, Pang Wei Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. 2023. Factscore: Fine-grained atomic evaluation of factual precision in long form text generation. *arXiv preprint arXiv:2305.14251*.
- Abhika Mishra, Akari Asai, Vidhisha Balachandran, Yizhong Wang, Graham Neubig, Yulia Tsvetkov, and Hannaneh Hajishirzi. 2024. [Fine-grained hallucination detection and editing for language models](#).
- Dor Muhlgay, Ori Ram, Inbal Magar, Yoav Levine, Nir Ratner, Yonatan Belinkov, Omri Abend, Kevin Leyton-Brown, Amnon Shashua, and Yoav Shoham. 2023. [Generating benchmarks for factuality evaluation of language models](#). In *Conference of the European Chapter of the Association for Computational Linguistics*.
- AI NIST. 2023. Artificial intelligence risk management framework (ai rmf 1.0).
- Artidoro Pagnoni, Vidhisha Balachandran, and Yulia Tsvetkov. 2021a. [Understanding factuality in abstractive summarization with frank: A benchmark for factuality metrics](#). *ArXiv*, abs/2104.13346.
- Artidoro Pagnoni, Vidhisha Balachandran, and Yulia Tsvetkov. 2021b. [Understanding factuality in abstractive summarization with FRANK: A benchmark for factuality metrics](#). In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 4812–4829, Online. Association for Computational Linguistics.

- Sida Peng, Eirini Kalliamvakou, Peter Cihon, and Mert Demirel. 2023. The impact of ai on developer productivity: Evidence from github copilot. *arXiv preprint arXiv:2302.06590*.
- Fabio Petroni, Tim Rocktäschel, Patrick Lewis, Anton Bakhtin, Yuxiang Wu, Alexander H. Miller, and Sebastian Riedel. 2019. [Language models as knowledge bases?](#) In *Conference on Empirical Methods in Natural Language Processing*.
- Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. 2020. [Exploring the limits of transfer learning with a unified text-to-text transformer](#). *Journal of Machine Learning Research*, 21(140):1–67.
- Inioluwa Deborah Raji, I Elizabeth Kumar, Aaron Horowitz, and Andrew Selbst. 2022. The fallacy of ai functionality. In *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency*, pages 959–972.
- Rohan Taori, Ishaan Gulrajani, Tianyi Zhang, Yann Dubois, Xuechen Li, Carlos Guestrin, Percy Liang, and Tatsunori B. Hashimoto. 2023. Alpaca: A strong, replicable instruction-following model. <https://crfm.stanford.edu/2023/03/13/alpaca.html>. Accessed: 6/15/2024.
- Together AI. 2023. Releasing 3b and 7b redpajama-incite family of models including base, instruction-tuned and chat models. <https://www.together.ai/blog/redpajama-models-v1>. Accessed: 6/15/2024.
- Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, et al. 2023a. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*.
- Hugo Touvron, Louis Martin, Kevin R. Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shrutu Bhosale, Daniel M. Bikel, Lukas Blecher, Cristian Cantón Ferrer, Moya Chen, Guillem Cucurull, David Esiobu, Jude Fernandes, Jeremy Fu, Wenyin Fu, Brian Fuller, Cynthia Gao, Vedanuj Goswami, Naman Goyal, Anthony S. Hartshorn, Saghar Hosseini, Rui Hou, Hakan Inan, Marcin Kardas, Viktor Kerkez, Madian Khabsa, Isabel M. Kloumann, A. V. Korenev, Punit Singh Koura, Marie-Anne Lachaux, Thibaut Lavril, Jenya Lee, Diana Liskovich, Yinghai Lu, Yuning Mao, Xavier Martinet, Todor Mihaylov, Pushkar Mishra, Igor Molybog, Yixin Nie, Andrew Poulton, Jeremy Reizenstein, Rashi Rungta, Kalyan Saladi, Alan Schelten, Ruan Silva, Eric Michael Smith, R. Subramanian, Xia Tan, Binh Tang, Ross Taylor, Adina Williams, Jian Xiang Kuan, Puxin Xu, Zhengxu Yan, Iliyan Zarov, Yuchen Zhang, Angela Fan, Melanie Kambadur, Sharan Narang, Aurelien Rodriguez, Robert Stojnic, Sergey Edunov, and Thomas Scialom. 2023b. [Llama 2: Open foundation and fine-tuned chat models](#). *ArXiv*, abs/2307.09288.
- Neeraj Varshney, Wenlin Yao, Hongming Zhang, Jian-shu Chen, and Dong Yu. 2023. [A stitch in time saves nine: Detecting and mitigating hallucinations of llms by validating low-confidence generation](#). *ArXiv*, abs/2307.03987.
- David Wadden, Kyle Lo, Bailey Kuehl, Arman Cohan, Iz Beltagy, Lucy Lu Wang, and Hannaneh Hajishirzi. 2022. Scifact-open: Towards open-domain scientific claim verification. *arXiv preprint arXiv:2210.13777*.
- Karen Weise and Cade Metz. 2023. When a.i. chatbots hallucinate. <https://www.nytimes.com/2023/05/01/business/ai-chatbots-hallucination.html>.
- Orion Weller, Marc Marone, Nathaniel Weir, Dawn J Lawrie, Daniel Khashabi, and Benjamin Van Durme. 2023. [“according to . . .”: Prompting language models improves quoting from pre-training data](#). In *Conference of the European Chapter of the Association for Computational Linguistics*.
- Muru Zhang, Ofir Press, William Merrill, Alisa Liu, and Noah A Smith. 2024. How language model hallucinations can snowball. *ICML*.
- Xingxing Zhang and Mirella Lapata. 2017. [Sentence simplification with deep reinforcement learning](#). In *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing*, pages 595–605. Association for Computational Linguistics.
- Yue Zhang, Yafu Li, Leyang Cui, Deng Cai, Lemao Liu, Tingchen Fu, Xinting Huang, Enbo Zhao, Yu Zhang, Yulong Chen, Longyue Wang, Anh Tuan Luu, Wei Bi, Freda Shi, and Shuming Shi. 2023. [Siren’s song in the ai ocean: A survey on hallucination in large language models](#). *ArXiv*, abs/2309.01219.
- Wenting Zhao, Tanya Goyal, Yu Ying Chiu, Liwei Jiang, Benjamin Newman, Abhilasha Ravichander, Khyathi Chandu, Ronan Le Bras, Claire Cardie, Yuntian Deng, et al. 2024a. Wildhallucinations: Evaluating long-form factuality in llms with real-world entity queries. *arXiv preprint arXiv:2407.17468*.
- Wenting Zhao, Xiang Ren, Jack Hessel, Claire Cardie, Yejin Choi, and Yuntian Deng. 2024b. Wildchat: 1m chatgpt interaction logs in the wild. *arXiv preprint arXiv:2405.01470*.

A Prompt Construction Details

We describe the process of constructing **HALO-GEN** . This benchmark consists of content-grounded tasks such as text summarization, as well as ungrounded text generation tasks. For ungrounded text generation, we focus on knowledge-oriented, rather than creative or subjective, tasks. We define a hallucination to be a fact in a model

generation that is not aligned with established world knowledge or with provided context. For content-grounded tasks, we consider hallucinations to be facts generated by a model that are not entailed by the provided context, even if factually correct.

It should be noted that there is no one definition of established knowledge for several facts, that truth can be pluralistic, and that data stores may contain conflicting information sources. We operationalize an ‘established’ knowledge source by specifying a singular ‘source of truth’ for each scenario, but it is possible for a practitioner to make different factuality determinations by considering different knowledge sources, or by interpreting information from the knowledge source differently.

Code Packages LLMs are frequently tasked with providing coding solutions (Zhao et al., 2024b; Peng et al., 2023). Prior work has noted that generative models can hallucinate code packages, and these hallucinations can present a security vulnerability (Bar Lanyado, 2023). This study measures the extent to which models hallucinate libraries in code generation scenarios. *Prompt Construction:* We obtain questions from Stack Overflow⁴, based on posts in 50 different subject areas we manually compiled. Subject areas we considered to source python programs included: Operating Systems, Architecture, Tree, Cloud, IoT (Internet of Things), Graph, OOP (Object-Oriented Programming), Optimization, DevOps, Unit Testing, Recursion, Blockchain, Bit Manipulation, Computer Vision, Security, Data Analysis, Amazon Web Services (AWS), Sorting, Dynamic Programming, Video Processing, Data Structures, Memory Management, Artificial Intelligence (AI), Exception Handling, Audio Processing, Web Scraping, Robotics, Quantum Computing, List, Augmented Reality (AR), Multithreading, Algorithm, Microsoft Azure, Machine Learning (ML), Virtual Reality (VR), Queue, Natural Language Processing (NLP), Serialization, Python, Math, Design Patterns, Web Frameworks, Regular Expressions (Regex), Stack, Parsing, Embedded Systems, Search, Google Cloud Platform (GCP), Hash, String.

We retained questions that contained the words ‘how to’, and were about the Python programming language.

⁴<https://stackoverflow.com/>

Summarization We study the extent to which LLMs hallucinate facts in summarization, a content-grounded task wherein a model is provided a piece of text and tasked with synthesizing the most salient information within that text. *Prompt Construction:* We extract 1300 randomly selected instances from the CNN/DailyMail dataset (Hermann et al., 2015), and include instructions as shown in Table 1. After filtering out duplicates, we are left with 1278 instances.

Simplification Text simplification is a content-grounded task wherein a model is provided a piece of text and is tasked with paraphrasing it in order to make the text easier to read and understand. *Prompt Construction:* For text simplification, we construct prompts from 1k instances sampled from the WikiLarge dataset (Zhang and Lapata, 2017), and include instructions as shown in Table 1.

Biographies This task measures the ability of language models to generate factually accurate statements about real people. *Prompt Construction:* We use the FactScore dataset (Min et al., 2023), which contains a total of 683 entities associated with corresponding Wikipedia articles. We worked with 682 entities, excluding the entity “Francisco Urroz.” Prompts are of the form “Tell me a bio of <entity>.”

Rationalization (Binary) The binary rationalization task measures the ability of language models to answer yes/no questions and provide justification based on the binary response. *Prompt Construction:* To create a dataset of prompts with Yes/No responses, we use three datasets requiring a model to generate a binary response along with a justification (Zhang et al., 2024). Each of these datasets are fixed with a specific label (either yes or no), and the tasks involve testing for primality, finding a senator who represented a specific state and attended a specific US college, and identifying if a flight sequence exists between any two cities.

- **Primality Testing:** This dataset consists of 500 randomly selected prime numbers falling within the range of 1000 to 20000. The correct response for each query is consistently “Yes” since all the provided numbers are prime. However, if the model provides an incorrect answer, it should provide an incorrect factorization as justification. Prompts are of the form “Is <number> a prime number? First, respond with yes or no. If no, then provide its factorization.”